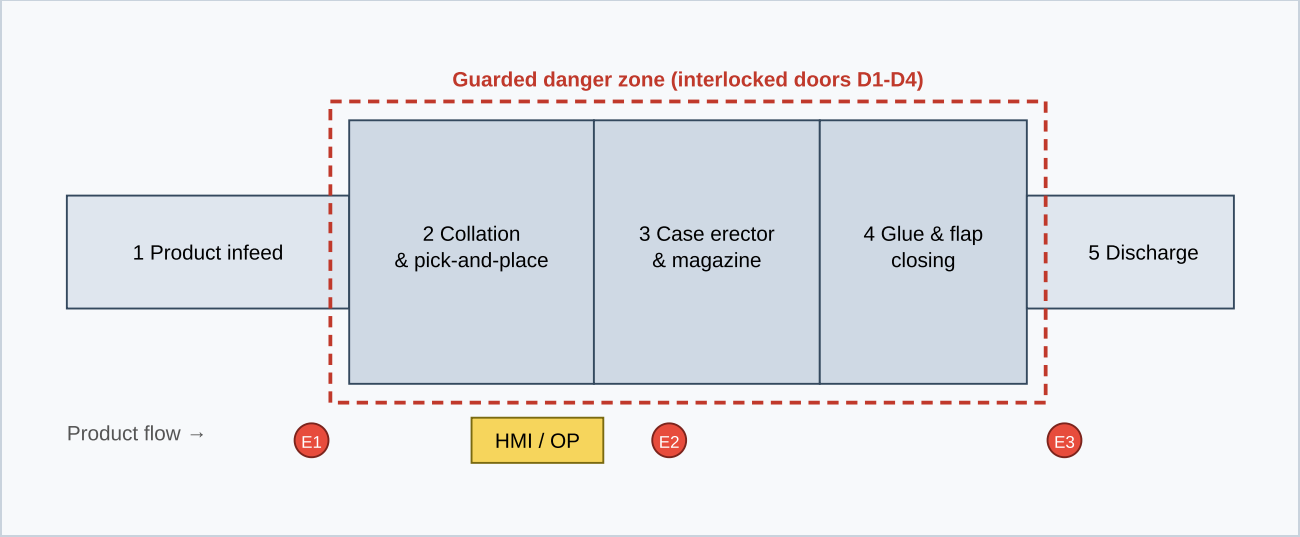
SAMPLE — example data

ACME PACKAGING MACHINERY CO. (FICTIONAL)

CP-200 Case Packer

Operating and Maintenance Manual



Model	CP-200
Serial number	_____
Document	OM-CP200-EN
Revision	C · October 2026
Language	English (original instructions)

⚠ Read and understand this manual before installing, operating or servicing the machine. Keep it with the machine.

This is a fictional sample created to show ManualOS output. The machine, company and all data are invented. Do not use for real equipment.

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0 About This Manual

0.1 Revision History

Revision	Date	Description	Approved
A	2026-03-02	First release	J. Ortega
B	2026-07-15	Added low-glue alarm and 2,000-hour service tasks	J. Ortega
C	2026-10-24	Updated spare parts list; new HMI screens	M. Patel

0.2 Purpose and Scope

This manual explains how to transport, install, commission, operate, maintain and decommission the Acme CP-200 Case Packer safely. It is written for operators, maintenance technicians and installers. Keep it next to the machine for its entire service life and pass it on if the machine is sold.

Related documents supplied with the machine: electrical schematics (drawing set E-200), pneumatic schematic (P-200), illustrated parts catalog, and the hot-melt unit manufacturer's manual. This manual is also available as a downloadable, printable PDF from the address on the machine nameplate.

0.3 Safety Messages and Symbols

Safety messages in this manual follow ANSI Z535.6 and ISO 3864-2. Each message names the hazard, explains what can happen, and tells you how to avoid it.

Signal word	Meaning
⚠ DANGER	A hazardous situation that **will** result in death or serious injury if not avoided.
⚠ WARNING	A hazardous situation that **could** result in death or serious injury if not avoided.
⚠ CAUTION	A hazardous situation that could result in minor or moderate injury if not avoided.
NOTICE	Information about practices that could damage the machine or property. Not related to personal injury.

In procedures, numbered steps must be done in order. Buttons on the operator panel are shown in **bold**; screen names are shown in *italics*.

1 Safety

1.1 General Safety Instructions

- Read this manual completely before working on the machine.
- Only trained and authorized persons may operate or service the machine.
- Never operate the machine with a guard open, removed or damaged.
- Do not modify the machine or its control program without written approval from the manufacturer.
- Report any damage, unusual noise or malfunction to your supervisor immediately.
- Follow all site safety rules in addition to this manual.

⚠ DANGER

Bypassing, removing or defeating interlocks, guards or safety functions.

Exposes people to all machine hazards and will lead to death or serious injury.

- Never bridge, remove or tamper with safety devices.
- Report defective safety devices and take the machine out of service until repaired.

1.2 Personnel Qualifications

Role	Qualification	Permitted tasks
Operator	Trained on this machine (Section 5.4)	Start, stop, load case blanks, refill glue, clear jams in the infeed, daily checks
Setup technician	Operator training plus changeover training	Format changeover, recipe selection, adjustments in Setup mode
Mechanical maintenance	Qualified mechanic with lockout training	Mechanical maintenance, part replacement, lubrication
Electrical maintenance	Qualified electrical worker (NFPA 70E)	Work in the electrical cabinet, drive and PLC replacement

1.3 Intended Use

The CP-200 is designed to erect corrugated regular slotted cases (RSC), load them with sealed packaged products (cartons, bottles or pouches) and close them with hot-melt adhesive. It is intended for indoor industrial use within the limits given in Chapter 12.

- Products: 50–1,500 g per unit, maximum 12 kg per case
- Case sizes: length 200–450 mm, width 150–350 mm, height 100–350 mm
- Speed: up to 20 cases per minute
- Environment: 5–40 °C, 20–85 % relative humidity, non-condensing, non-explosive atmosphere

1.4 Foreseeable Misuse

The following uses are prohibited:

- Packing unsealed food, loose liquids, or hazardous goods
- Operating with guard doors open or interlocks bypassed
- Climbing on or riding the machine or its conveyors
- Using adhesives other than those specified in Section 9.3
- Operation in potentially explosive atmospheres
- Using the machine as a work platform or for storage

1.5 Danger Zones and Operator Positions

The danger zone is the fully guarded area from the end of the infeed conveyor to the case discharge. During production, the operator works only at the operator panel and at the case-blank magazine loading position outside the guards.

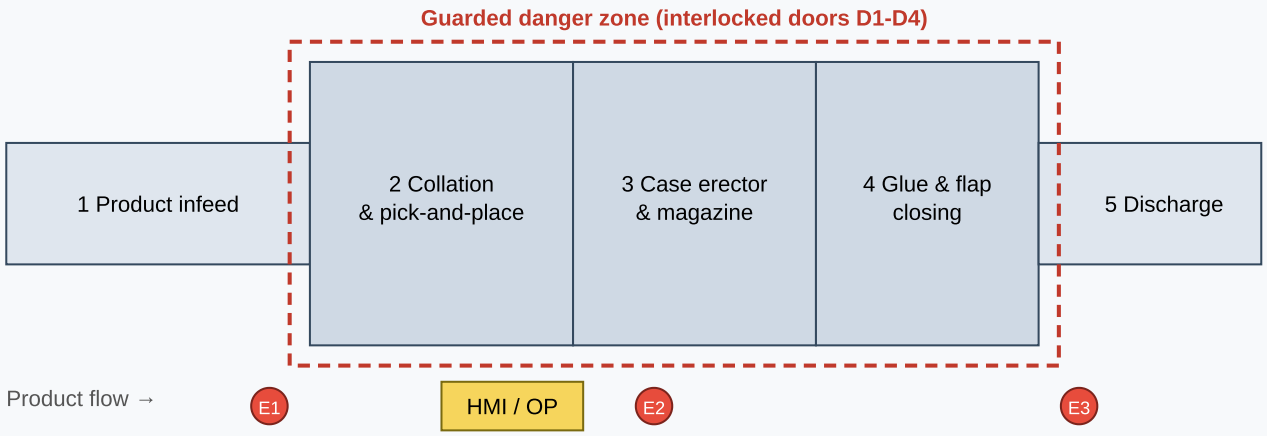


Figure 1-1. Plan view: machine modules, guarded danger zone, operator panel and emergency stop locations (E1–E3).

1.6 Safety Devices

Device	Location	Function	Check
Emergency stop buttons E1–E3	Infeed, operator panel, discharge	Stops all motion (Category 1 stop), removes power to drives and air	Daily
Guard door interlocks D1–D4	All four access doors	Opening a door stops motion; restart only after door closed and reset	Daily
Light curtain LC1	Case blank magazine	Stops the erector if the operator reaches into the magazine	Weekly
Safe Torque Off	Servo and conveyor drives	Removes motor torque on E-stop or door open	Yearly test

Device	Location	Function	Check
Pneumatic dump valve	Air inlet module	Exhausts machine air on E-stop and at lockout	Weekly
Stack light	Top of operator panel	Red = stopped/fault, Amber = warning, Green = running, Blue = operator action needed	Daily

1.7 Emergency Stop

Press any red emergency stop button to stop the machine immediately. All motion stops, servo torque is removed and the pneumatic system is exhausted. The hot-melt unit stays powered so the glue does not solidify.

To restart: remove the cause of the emergency, pull out the emergency stop button, press **Reset** on the operator panel, then press **Start**. Releasing the button alone never restarts the machine.

1.8 Safety Labels

Label	Location	Meaning
Electrical hazard	Electrical cabinet door	Hazardous voltage inside; qualified persons only
Hot surface	Glue tank and glue heads	Burn hazard; let cool before touching
Crushing of hands	Case erector and flap closer	Keep hands clear of moving parts
Entanglement	Infeed belt end rollers	Keep guards closed; no loose clothing
Read the manual	Operator panel	Read manual before operation

Replace any label that is damaged or unreadable. Part numbers are listed in Chapter 10.

1.9 Personal Protective Equipment

Task	Eyes	Hands	Feet	Other
Production	Safety glasses	—	Safety shoes	Hearing protection if above 80 dB(A)
Glue refill / glue head cleaning	Safety glasses	Heat-resistant gloves	Safety shoes	Long sleeves
Format changeover	Safety glasses	Cut-resistant gloves	Safety shoes	—
Electrical work	Face shield	Voltage-rated gloves	Safety shoes	Arc-rated clothing as labeled

1.10 Lockout/Tagout

Lock out the machine before any cleaning, maintenance or jam clearing inside the guards. The CP-200 has these energy sources:

Energy	Isolation point	Release stored energy	Verify
Electrical, 480 V 3-phase	Main disconnect Q0 on cabinet door	Wait 5 min for drive capacitors to discharge	Try Start; measure at drive terminals
Compressed air, 6 bar	Lockable ball valve V0 at air inlet	Dump valve exhausts automatically; check gauge	Gauge reads 0 bar
Gravity	—	Lower the pick-and-place head to its rest position	Visual check
Heat (hot-melt)	Hot-melt unit disconnect Q5	Allow 30 min to cool	Surface below 50 °C

1. Tell everyone affected that the machine will be locked out.
2. Stop the machine with the **Stop** button.
3. Open main disconnect Q0, close air valve V0 and (if needed) hot-melt disconnect Q5.
4. Attach your personal lock and tag to each isolation point.
5. Release or block stored energy as shown in the table.
6. Verify zero energy: press **Start**, check the air gauge, measure voltage.
7. Do the work.
8. When finished: remove tools, refit guards, make sure everyone is clear, remove locks and tell affected people.

⚠ DANGER

Drive and power-supply capacitors stay charged after power is removed.

Touching DC bus terminals before discharge will cause death or severe injury.

- After isolation wait at least the stated discharge time (5 min).
- Measure the DC bus before work and confirm it is below 50 V.

1.11 Residual Risks

The following risks remain despite the guards and safety functions. Observe the instructions in each message.

⚠ DANGER

Hazardous voltage inside electrical enclosures and terminal boxes.

Contact with live parts will cause death or severe burns.

- Only qualified electrical workers may open the enclosure.
- Switch off and lock the main disconnect before opening.
- Verify absence of voltage with a rated tester on every conductor before touching.

⚠ WARNING

Crushing points at the case erector, pick-and-place head and flap closer.

Body parts caught in the gap can be crushed or amputated.

- Keep hands and body clear of the motion path while the machine is energised.
- Use setup mode with reduced speed and an enabling device for close-in work.
- Block or lower gravity-loaded parts before reaching in.

⚠ WARNING

Rotating rollers and belts at the infeed and discharge conveyor ends.

Hair, clothing, gloves or jewellery can be drawn in, causing severe injury or death.

- Keep guards closed while parts rotate.
- Tie back long hair; no loose clothing, jewellery or gloves near rotating parts.
- Wait for complete standstill before opening guards.

⚠ WARNING

Hot-melt tank, hoses and glue heads operate at up to 180 °C.

Touching hot parts or molten glue causes severe burns.

- Wear heat-resistant gloves and safety glasses when refilling glue or cleaning nozzles.
- Let the glue unit cool for 30 minutes before maintenance.

⚠ WARNING

Compressed air (6 bar) in the vacuum, erector and flap-folding cylinders.

Sudden movement of cylinders, whipping hoses or flying parts can cause injury.

- Close and lock the air isolation valve and exhaust residual air before work.
- Secure loads held by cylinders.
- Never point compressed air at people.

⚠ CAUTION

Change parts and bundles of case blanks can weigh more than 23 kg.

Can cause back and muscle injuries.

- Use lifting aids for loads above 23 kg (or local limit).
- Lift with two people where aids are not available.

1.12 Noise and Emissions

The A-weighted emission sound pressure level at the operator position is 74 dB(A) (uncertainty K = 3 dB), measured to ISO 11204 at 20 cases/min with 300 × 250 × 200 mm cases. The peak C-weighted value does not exceed 130 dB. Hot-melt adhesive can release small amounts of fumes; provide general room ventilation.

1.13 Network Security

The PLC and operator panel connect to the plant network through a managed switch. Remote access is disabled by default. Operator panel functions are protected by four user levels (Operator, Setup, Maintenance, Engineer). Change the default passwords during commissioning.

SAMPLE — example data · fictional machine

2 Machine Description and Theory of Operation

2.1 What the Machine Does

The CP-200 takes a steady stream of sealed products and turns it into closed, glued shipping cases ready for the pallet. Flat corrugated blanks go in at one side, products come in from the upstream line, and a finished, square, sealed case leaves at the other end, up to 20 times a minute.

Every case passes through five stations in a fixed order: the **infeed conveyor** spaces and counts the products, the **collation and pick-and-place** station groups them into the case pattern and lowers them into an open case, the **case erector** forms each case from a flat blank, the **glue and flap closing** station seals it, and the **discharge conveyor** hands it on to the downstream line. Sections 2.4 to 2.8 describe how each station works; Section 2.9 describes how they work together in one cycle.

2.2 Machine Layout and Product Flow

Products flow from left to right as seen from the operator panel. Stations 2 to 4 are inside the interlocked guard; the infeed and discharge conveyors pass through tunnel openings in the guard.

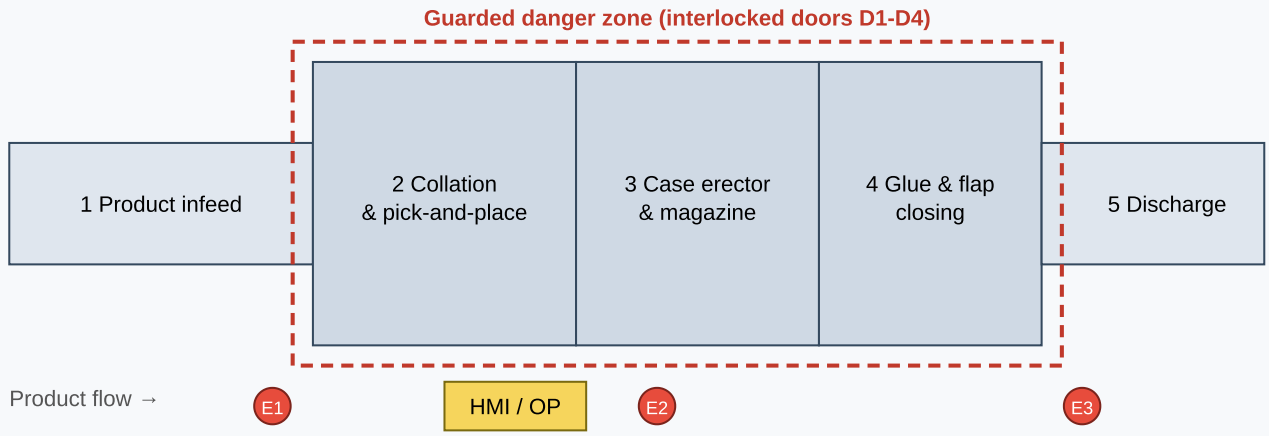


Figure 2-1. Machine layout in product-flow order. Numbers match the station sections below.

2.3 Main Assemblies and Their Functions

No.	Assembly	What it does	Section
1	Infeed conveyor	Brings products in single file, spaces and counts them	2.4
2	Collation and pick-and-place	Groups products into the case pattern and loads them into the open case	2.5
3	Case erector and blank magazine	Stores flat blanks and forms one open case per cycle	2.6
4	Glue and flap closing	Applies hot-melt adhesive, folds and presses the flaps shut	2.7

No.	Assembly	What it does	Section
5	Case discharge conveyor	Moves the sealed case out and hands it to the downstream line	2.8
6	Electrical cabinet	Controller, drives, safety relay, power distribution	—
7	Operator panel	Touch screen and Start / Stop / Reset buttons, emergency stop E2	Ch. 6

2.4 How the Infeed Conveyor Works

The infeed conveyor is the machine's entry point. It takes products from the upstream line, opens up a regular gap between them so they can be counted reliably, and holds products back whenever the collation station cannot accept more.

Main components	Flat-top belt conveyor (2,350 mm) driven by a gearmotor on a variable-speed drive; adjustable side guides; metering belt at the discharge end
How it moves	Main belt runs continuously at line speed; the metering belt runs about 15 % faster to open a gap between products
Sensors	Product sensor at the metering belt counts each product; backup sensor 1 m upstream detects a queue forming
Interlocks / permissives	Runs only in Automatic with the machine running and air pressure OK; stops when the collation station is full or on any red alarm
Operator interaction	Set side-guide width at changeover (color-coded scale); clear fallen products at the tunnel opening
Program reference	Infeed logic, rungs 0–13

When the machine starts, the main belt runs first; the metering belt starts once the collation station reports that its lane is clear. Each product breaking the product sensor beam adds one to the count for the current case. If the backup sensor stays blocked for 3 seconds, the infeed slows to half speed; after 10 seconds it stops and shows alarm A110. **Inferred – please confirm** *Belt speed ratio inferred from CAD pulley sizes.*

⚠ WARNING

Rotating rollers and belts at the infeed conveyor ends and the metering transfer.

Hair, clothing, gloves or jewellery can be drawn in, causing severe injury or death.

- Keep guards closed while parts rotate.
- Tie back long hair; no loose clothing, jewellery or gloves near rotating parts.
- Wait for complete standstill before opening guards.

2.5 How the Collation and Pick-and-Place Works

This station builds the product pattern for one case (for example 3 rows × 4 products) and places the complete layer into the open case in one move. It is the heart of the machine and sets the overall cycle rate.

Main components	Lane divider and collation fingers; two-axis servo gantry (horizontal and vertical) with a vacuum pick head; vacuum generator with pressure switch
How it moves	Collation fingers index one row at a time; the gantry travels 600 mm horizontally and 350 mm vertically; the pick head lowers, grips by vacuum, lifts, travels, and lowers into the case
Sensors	Row-complete sensor; vacuum pressure switch confirms grip; servo position feedback on both axes
Interlocks / permissives	Pick only when a full pattern is collated; place only when the case erector reports a case open and in position; head returns to the safe height before horizontal travel
Operator interaction	Change collation fingers at format changeover; jog axes in Setup mode from the *Manual Control* screen
Program reference	Pick-and-place logic, rungs 0–27

Products are pushed into lanes and stopped by the collation fingers until each row is complete. When the last row arrives, the pick head descends, switches on vacuum and waits for the pressure switch to confirm a grip before lifting. It then travels to the case, lowers the pattern to just above the case bottom and releases. If the grip is lost during travel, the head stops, lowers the products back onto the collation plate and raises alarm A130. The vertical axis has a holding brake, so the head stays in place on power loss.

⚠ WARNING

The pick-and-place head is held up by a motor brake and can drop if the brake is released or fails.

A falling load can crush or kill.

- Mechanically block or lower vertical axes before maintenance.
- Never stand under suspended loads.
- Test holding brakes at the specified interval.

2.6 How the Case Erector Works

The case erector forms a square, open-topped case from a flat corrugated blank, ready for loading. One case is erected for every product pattern.

Main components	Gravity-fed blank magazine (up to 150 blanks); swing arm with vacuum cups; forming plow and bottom-flap folders driven by pneumatic cylinders
How it moves	Swing arm pulls a blank from the magazine and rotates it 90°; forming plow opens the blank into a tube; folders close the minor and major bottom flaps
Sensors	Blank-present and magazine-low sensors; vacuum switch on the swing arm; case-square sensor in the loading position; light curtain LC1 at the magazine
Interlocks / permissives	Starts a new case only when the previous case has left the loading position; stops immediately if the light curtain is interrupted
Operator interaction	Reload blanks from outside the guard while running; adjust magazine guides at changeover
Program reference	Case erector logic, rungs 0–21

The swing arm presses its vacuum cups against the front blank, waits for the vacuum switch, then rotates. As it swings, the blank meets the forming plow, which opens it into a square tube. Two cylinders fold the minor flaps and a third pushes the major flaps into position, so the case bottom is closed and glued in the next station. If no vacuum is reached within 1.5 seconds, the arm retries once and then raises alarm A130. **Inferred – please confirm** *Bottom flaps glued at station 4, inferred from CAD glue-gun position — confirm.*

⚠ WARNING

Crushing points between the swing arm, forming plow and flap folders of the case erector.

Body parts caught in the gap can be crushed or amputated.

- Keep hands and body clear of the motion path while the machine is energised.
- Use setup mode with reduced speed and an enabling device for close-in work.
- Block or lower gravity-loaded parts before reaching in.

2.7 How the Glue and Flap Closing Works

This station seals the case. It applies hot-melt adhesive to the flaps, folds the top flaps closed and holds the case under pressure until the glue sets.

Main components	Hot-melt unit with 10 kg tank, heated hoses and four glue guns; top-flap plows; compression belts on both sides
How it moves	Case moves through on the compression belts; fixed plows fold the leading, trailing and side flaps; glue guns fire as each flap passes
Sensors	Case-entry sensor triggers the glue pattern; encoder on the compression belt times each glue dot; glue-ready signal from the hot-melt unit
Interlocks / permissives	Cases are not released into this station unless the glue is at temperature; glue guns fire only while a case is moving past them
Operator interaction	Refill glue (amber alarm A141); check glue dots on the first cases after a restart
Program reference	Glue and flap logic, rungs 0–19

When the case-entry sensor sees the leading edge, the control counts encoder pulses and fires each glue gun at the programmed distance, so the glue pattern stays in the same place on the flap at any speed. The plows fold the flaps onto the glue, and the compression belts hold the case square for about 1.5 seconds while the adhesive sets. If the glue temperature drops, the station finishes the case in progress and then holds new cases until the glue recovers.

⚠ WARNING

Hot-melt tank, hoses and glue guns operate at up to 180 °C.

Touching hot parts or molten glue causes severe burns.

- Wear heat-resistant gloves and safety glasses when refilling glue or cleaning nozzles.
- Let the glue unit cool for 30 minutes before maintenance.

2.8 How the Case Discharge Works

The discharge conveyor carries the sealed case out of the guard and hands it to the downstream line, and it tells the CP-200 when the downstream line is backing up.

Main components	Powered roller conveyor (1,200 mm) with a gearmotor
How it moves	Runs continuously while cases are present; stops 30 seconds after the last case leaves
Sensors	Case-exit sensor; downstream-full signal from the next machine
Interlocks / permissives	When the downstream line is full, the conveyor stops and the machine finishes the case in progress, then pauses
Operator interaction	None in normal production; clear the downstream conveyor if alarm A160 appears
Program reference	Discharge logic, rungs 0–9

The machine restarts by itself when the downstream-full signal clears, so the operator does not need to press Start after a short downstream stop.

2.9 Sequence of Operation

One automatic cycle, from the product's point of view. Stations work in parallel: while one case is being loaded, the next blank is already being erected and the previous case is being sealed.

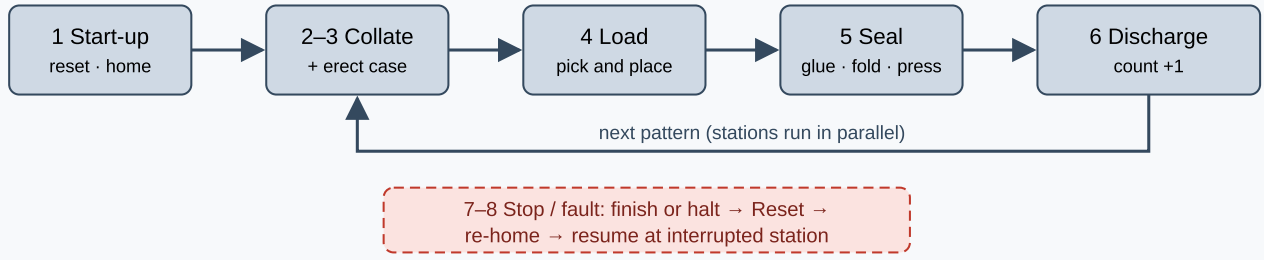


Figure 2-2. Automatic cycle. The dashed path shows what happens after a stop or fault.

1. **Start-up.** After **Reset** and **Start**, the gantry moves to its home position, the case erector makes the first case, and the infeed belts start.
2. **Collate.** The infeed spaces and counts products; the collation fingers build rows until one full pattern is ready.
3. **Erect.** In parallel, the swing arm pulls a blank, forms it into a case and folds the bottom flaps. The case waits in the loading position.
4. **Load.** With a full pattern and an open case both ready, the pick head grips the pattern by vacuum, travels and lowers it into the case.
5. **Seal.** The loaded case is pushed onto the compression belts; glue is applied and the flaps are folded and pressed shut.
6. **Discharge.** The sealed case leaves on the discharge conveyor and the case count increases by one.
7. **Cycle stop.** Pressing **Stop** lets the current pattern finish loading and the last case clear the sealer, then all stations stop in their home positions.
8. **After a fault or E-stop.** Motion stops immediately. After the cause is removed and **Reset** is pressed, the machine re-homes the gantry and resumes from the station that was interrupted.

2.10 Interlocks and Machine States

Condition	What it does	Stations affected	How to clear
Guard doors open or E-stop pressed	Stops all motion; machine cannot start	All	Close doors / release E-stop, press Reset
Air pressure below 5 bar	Prevents start; stops after current cycle	Erector, pick-and-place	Restore plant air
Glue not at temperature	Holds new cases before the sealer	Glue and flap closing	Wait for heat-up
No open case in loading position	Pick head waits with the pattern	Pick-and-place	Automatic; check erector if it persists
Collation full	Infeed metering belt pauses	Infeed	Automatic
Downstream line full	Discharge stops; machine pauses after current case	Discharge, then all	Automatic when downstream clears
Magazine below 20 blanks	Amber warning only	Case erector	Reload blanks

Machine states shown on the *Home* screen: **Stopped** → **Ready** (reset, all permissives met) → **Running** → **Cycle stopping** → **Stopped**; any red alarm moves the machine to **Faulted** until reset. Safety functions are described in Section 1.6.

2.11 Control Logic Description

This section explains what the control program makes the machine do, grouped by machine function. Each group summarizes several related rungs of the program; the small reference shows which rungs. All 127 rungs of the program are accounted for, as listed in Appendix 12.3.

Modes, starting and stopping

Function	What the logic does	Rungs
Mode selection	Reads the key switch and allows Automatic only with all doors closed; Setup mode limits every axis to jog speed and enables the hold-to-run button.	0–3
Ready, start and cycle stop	Combines the start permissives (safety reset, air OK, glue ready, all axes homed) into a single Ready state; **Start** latches Running, **Stop** requests a cycle stop that ends when every station is back at home.	4–11
Homing after reset	Moves the gantry up first, then across, before any other station may start.	12–14
Stack light and counters	Drives the stack-light colors from the machine state and counts finished cases and shift totals.	15–17

Infeed conveyor

Function	What the logic does	Rungs
Running the infeed	Runs the main belt while the machine is running; starts the metering belt only when the collation lane is clear.	0–5
Counting products	Counts each product passing the metering sensor toward the current pattern and resets the count when the pattern is picked.	6–9
Backup protection	Slows the infeed after 3 s of queue at the backup sensor and stops it with alarm A110 after 10 s.	10–13

Collation and pick-and-place

Function	What the logic does	Rungs
Building the pattern	Indexes the collation fingers one row at a time and signals Pattern Ready when the last row is complete.	0–7
Pick, travel and place	Runs the pick-place-return motion as a step sequence: lower, vacuum on, confirm grip, lift, travel, lower into case, release, return. Each step waits for the axis to arrive before the next begins.	8–21
Grip-loss recovery	If vacuum is lost during travel, stops the gantry, sets the pattern back down and raises alarm A130.	22–24
Brake and following-error monitoring	Holds the vertical brake applied whenever the axis is not enabled; raises A120 if an axis lags its commanded position.	25–27

SAMPLE — example data · fictional machine

Case erector

Function	What the logic does	Rungs
Picking a blank	Starts a new case once the loading position is empty; waits up to 1.5 s for vacuum and retries once.	0–8
Forming and bottom-flap folding	Times the minor- and major-flap cylinders to the swing-arm position and confirms the case is square before reporting Case Open.	9–17
Magazine and light curtain	Raises the magazine-low warning at 20 blanks and stops the erector while the light curtain is interrupted.	18–21

Glue and flap closing

Function	What the logic does	Rungs
Glue readiness	Holds new cases before the station until the hot-melt unit reports temperature.	0–3
Glue pattern timing	Starts a glue pattern at the case leading edge and fires each gun at its programmed distance using the belt encoder, so dots stay in place at any speed.	4–15
Compression and release	Keeps the compression belts running until the case has been under pressure for 1.5 s, then releases it to discharge.	16–19

Case discharge

Function	What the logic does	Rungs
Discharge conveyor control	Runs while cases are present and for 30 s after; stops and pauses the machine when the downstream line is full, then resumes automatically.	0–9

Alarms and diagnostics

Function	What the logic does	Rungs
Alarm detection and reset	Turns each fault condition into the alarm codes in Section 8.1, latches red alarms until **Reset**, and lets amber alarms clear by themselves.	0–12

2.12 Nameplate

The nameplate is on the electrical cabinet door. It shows: manufacturer, model, serial number, year of manufacture, supply voltage and frequency, full-load current, short-circuit current rating, air pressure, machine mass, and compliance marks. Quote the model and serial number when contacting service.

2.13 Options

- Tamper-evident tape sealer instead of hot-melt (option T): replaces the glue guns with a top-and-bottom tape head; the sealing step is otherwise the same
- Case printer / label applicator interface (option L): prints or labels each case on the discharge conveyor

- Remote diagnostics gateway (option R)

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3 Transport, Handling and Storage

3.1 Transport and Lifting

The machine ships in two sections on skids. Lift only with a forklift of adequate capacity from the marked side, or with slings attached to the four lifting eyes.

Section	Mass	Dimensions (L × W × H)	Lifting
A – Infeed and collation	1,150 kg	2,400 × 1,500 × 2,100 mm	Forklift from side, or 4 lifting eyes
B – Erector, glue and discharge	1,450 kg	2,900 × 1,600 × 2,100 mm	Forklift from side, or 4 lifting eyes

⚠ WARNING

Heavy machine or modules during lifting and transport, including off-centre centre of gravity.

Tip-over or a dropped load can cause death or serious injury and damage the machine.

- Use only the marked lifting points and lifting gear rated for the stated mass.
- Fit transport locks before moving.
- Keep people out of the lift area.

3.2 Unpacking

Check the delivery against the packing list and inspect for transport damage before signing the delivery note. Report damage to the carrier and to the manufacturer within 48 hours. Remove the red transport locks only after the machine is in its final position.

3.3 Storage

Store indoors at 0–45 °C and below 85 % relative humidity, protected from dust. If stored for more than 6 months, carry out the checks in Section 5.3 before startup.

4 Installation

4.1 Site Requirements

- Level concrete floor, minimum load 750 kg/m², flatness ±3 mm over the machine length
- Service clearance of 1,000 mm on the operator side and 800 mm on all other sides
- Lighting of at least 500 lux at the operator panel

4.2 Placing and Leveling

1. Position section A and B in line, using the alignment pins at the joint.
2. Bolt the sections together with the M12 bolts supplied (tighten to 80 N·m).
3. Level each section with the adjustable feet to within 0.5 mm/m.
4. Anchor the feet to the floor with the brackets supplied.
5. Remove all red transport locks and keep them for future moves.

4.3 Utility Connections

Utility	Requirement
Electrical supply	480 V AC ±10 %, 3-phase + PE, 60 Hz; full-load current 32 A; SCCR 65 kA
Supply protection	40 A Class J fuses or equivalent circuit breaker
Compressed air	6 bar (87 psi), 350 l/min, ISO 8573-1 class 4.4.4, ½-inch connection
Network	Ethernet RJ45 at the cabinet top for plant network

NOTICE

Wrong supply voltage, phase rotation, air quality, lubricant or fluid.

Can damage the machine and void the warranty.

- Check supply data against the nameplate before connecting.
- Use only the specified lubricants and fluids.

4.4 Interface to Upstream and Downstream Equipment

The CP-200 exchanges *Ready*, *Running* and *Downstream Full* signals with adjacent machines through terminal strip X10. A line-wide emergency stop can be linked through safety terminal X11.

5 Commissioning

5.1 Pre-Start Checks

- All transport locks removed; all guards fitted and closed
- Supply voltage and phase rotation correct (conveyors run in the direction of the arrows)
- Air pressure 6 bar at the regulator
- Glue tank filled and at temperature
- Each emergency stop and door interlock tested and recorded

5.2 First Start-Up

1. Switch on main disconnect Q0. The operator panel starts in about 60 seconds.
2. Log in at the Setup level.
3. Press **Home** to move all axes to their reference positions.
4. Select a recipe and run 10 dry cycles without product.
5. Run 20 cases with product, inspect them and adjust as needed.
6. Record the results in the commissioning report.

5.3 Restarting After a Long Shutdown

After more than 6 months out of service, or after relocation, check belts and vacuum cups for aging, drain the air filter, replace the glue, and repeat Section 5.1.

5.4 Operator Training

Each operator needs a 4-hour training session covering this manual's Chapters 1, 6, 7 and 8, including a practical E-stop and jam-clearing exercise. Record training on the sign-off sheet.

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6 Controls and Operator Panel

6.1 Control Elements

Control	Type	Function
Start	Green illuminated button	Starts automatic production
Stop	Black button	Stops at the end of the current cycle
Reset	Blue illuminated button	Resets faults and safety circuit after the cause is removed; lit when reset is needed
Emergency stop E1–E3	Red mushroom button	Immediate stop of all motion
Mode selector	Key switch	Selects Automatic or Setup mode
Main disconnect Q0	Rotary handle, lockable	Switches the machine supply on and off

6.2 Operator Panel Screens

Screen	Purpose	User level
Home	Machine status, speed, case count, active alarms	All
Production	Start/stop, speed setting, counters	Operator
Recipes	Select or edit case and product formats	Setup
Manual Control	Jog individual axes in Setup mode	Setup
Alarms	Active alarms and alarm history	All
Maintenance	Service hour counters and maintenance reminders	Maintenance

7 Operation

7.1 Operating Modes

Mode	How to select	What happens
Automatic	Key switch to AUTO, all doors closed	Full-speed production; all guards active
Setup	Key switch to SETUP	Single-axis jog at reduced speed (max. 10 mm/s) with the hold-to-run button; door D2 may be open with the enabling device
Clean	Setup mode + Clean button on Maintenance screen	Conveyors run slowly for belt cleaning; all other motion locked

⚠ WARNING

Safeguards partly muted in setup, teach or maintenance mode.

Moving parts can reach the operator.

- Use the enabling device; release it to stop motion.
- Only one person in the danger zone controls motion.
- Return to automatic only after everyone has left the zone.

7.2 Daily Start-Up

1. Check that all guards are closed and nobody is inside the guarded area.
2. Switch on main disconnect Q0 and open air valve V0.
3. Wait for the glue to reach temperature (green light on the glue unit, about 20 minutes).
4. Load case blanks into the magazine.
5. Press **Reset**, then select the recipe on the *Recipes* screen.
6. Press **Start**. The stack light turns green.

7.3 During Production

- Keep the blank magazine filled; a low-blank warning appears at 20 blanks.
- Watch the first cases after any restart for correct glue and flap closing.
- Respond to blue stack light (operator action needed) and alarms promptly.

7.4 Format Changeover

Changeover takes about 15 minutes. Select the new recipe, then follow the on-screen checklist: change the collation fingers and magazine guides (color-coded for each format), adjust the flap-closer width with the handwheel to the value shown, and run 5 test cases.

⚠ WARNING

Sharp tools, blades, knives and cut edges.

Contact can cause deep cuts or amputation.

- Handle tools and blades with cut-resistant gloves.
- Use tool holders or guards when changing blades.
- Never reach past a guard to remove scrap.

7.5 Stopping

Stop	Use	Restart
Cycle stop (**Stop**)	Normal end of production	Press Start
Door open	Access for minor jam clearing	Close door, press Reset, Start
Emergency stop	Danger to people or machine	Release button, Reset, Start
End of shift	Production finished	Stop, turn off glue unit (or set to standby), switch off Q0

7.6 Cleaning

Clean the machine at the end of each shift: remove dust and carton scraps with a vacuum (not compressed air), and wipe the photo-sensor lenses with a soft cloth. Clean the glue nozzles only when hot, using the tools from the glue unit kit.

8 Alarms and Troubleshooting

8.1 Alarm Messages

Alarms appear on the *Alarms* screen with a colored banner. Red alarms stop the machine; amber alarms are warnings that let production continue. Correct the cause, then press **Reset**.

Code	Message	Type	Probable cause	Remedy
A101	Emergency stop pressed	Red	An E-stop button is pressed	Remove the cause, release the button, press Reset
A102	Guard door open	Red	Door D1–D4 opened during operation	Close the door, press Reset
A110	Infeed conveyor drive fault	Red	Motor overload or product jam on the infeed	Lock out, clear jam, check belt tension; reset drive
A120	Pick-and-place servo following error	Red	Axis blocked or mechanical binding	Check for obstruction in Setup mode; check belt and couplings
A130	Case blank not picked	Red	Vacuum cups worn or blank warped	Check vacuum cups and blanks; clean or replace cups
A140	Glue temperature low	Amber	Glue tank or hose below set point	Wait for heat-up; check heater and hose connection
A141	Glue level low	Amber	Glue tank below minimum	Refill with specified hot-melt (Section 9.3)
A150	Air pressure low	Red	Supply below 5 bar	Check plant air supply and filter regulator
A160	Downstream conveyor full	Amber	Discharge blocked by downstream machine	Clear downstream conveyor; machine restarts automatically
A170	Magazine low	Amber	Fewer than 20 blanks left	Reload the magazine
A180	Light curtain interrupted	Red	Reaching into the magazine while the erector runs	Withdraw, press Reset

8.2 Troubleshooting

Symptom	Probable cause	Remedy
Cases open after closing	Glue too cold or too little glue	Check glue temperature (175 °C); clean nozzles; increase dot length
Products fall during transfer	Wrong recipe or worn collation fingers	Check selected recipe; replace fingers
Case corners not square	Erector guides out of adjustment	Readjust guides to recipe values
Machine will not start, no alarm	Mode switch in Setup or Reset not pressed	Turn switch to AUTO, press Reset

Symptom	Probable cause	Remedy
Frequent blank misfeeds	Humid or warped blanks	Store blanks in a dry place; adjust magazine pressure

8.3 Emergencies and Jam Clearing

If a person is trapped: press the nearest emergency stop. The pick-and-place head can be moved by hand after pressing the brake release button inside door D2 (support the head first, as it will drop slowly). Call first aid.

To clear a jam inside the guards: stop the machine, lock out as described in Section 1.10, remove the product, check for damage, and restart as described in Section 7.2.

Glue fire: switch off the glue unit and use a CO_2 or dry-powder extinguisher. Never use water on hot glue.

9 Maintenance

9.1 Maintenance Safety

Lock out the machine before any maintenance task (Section 1.10). Only the personnel listed in Section 1.2 may carry out these tasks. After maintenance, refit all guards and test the safety devices before returning the machine to production.

⚠ DANGER

Hazardous voltage inside electrical enclosures and terminal boxes.

Contact with live parts will cause death or severe burns.

- Only qualified electrical workers may open the enclosure.
- Switch off and lock the main disconnect before opening.
- Verify absence of voltage with a rated tester on every conductor before touching.

9.2 Preventive Maintenance Schedule

Interval	Component	Task	By
Daily	Safety devices	Test each E-stop and door interlock	Operator
Daily	Photo sensors	Clean lenses	Operator
Daily	Glue nozzles	Check for drips and blockage	Operator
Weekly	Vacuum cups	Inspect for cracks and wear; replace if damaged	Maintenance
Weekly	Air filter regulator	Drain condensate; check 6 bar setting	Maintenance
Weekly	Light curtain	Test with test rod	Maintenance
Monthly / 500 h	Conveyor belts	Check tension and tracking	Maintenance
Monthly / 500 h	Linear guides	Lubricate (Section 9.3)	Maintenance
6-monthly / 2,000 h	Pick-and-place timing belt	Check tension; replace if worn	Maintenance
6-monthly / 2,000 h	Glue filter	Replace	Maintenance
Yearly	Safe Torque Off and safety circuit	Full functional test and record	Electrical
Yearly	Electrical cabinet	Check terminals for tightness; clean fan filters	Electrical

9.3 Lubrication and Fluids

Point	Lubricant / fluid	Quantity	Interval
Linear guides (8 points)	NLGI 2 lithium grease (food-grade H1 if required)	2 strokes per nipple	500 h
Gearbox, infeed conveyor	ISO VG 220 synthetic gear oil	0.4 l (lifetime fill)	Check yearly
Chain, flap closer	Chain spray, light oil	Thin film	Monthly
Hot-melt adhesive	EVA hot-melt, 160–180 °C application	Tank 10 kg	As needed

9.4 Inspecting Safety Functions

Test each safety device at the interval in Section 9.2. A device passes if the machine stops and cannot be restarted until the device is reset. Record the result in the maintenance log. Take the machine out of service if any test fails.

9.5 Replacing Vacuum Cups

1. Lock out the machine (Section 1.10).
2. Unscrew the worn cups from the erector suction bar by hand.
3. Screw in new cups (part no. CP2-1104) hand-tight.
4. Restore power, run 5 cycles in Setup mode and check pick-up.

9.6 Adjustment and Calibration

After replacing a servo motor or timing belt, re-reference the pick-and-place axis: in Setup mode, open the *Manual Control* screen, select the axis and press **Set Home** with the head at the marked reference position.

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10 Spare Parts

10.1 Recommended Spare Parts

Part no.	Description	Qty in machine	Recommended stock	Class
CP2-1104	Vacuum cup, 40 mm, nitrile	8	16	Wear
CP2-1210	Timing belt, pick-and-place, HTD 8M	1	1	Wear
CP2-1305	Infeed conveyor belt, 2,350 mm	1	1	Wear
CP2-1420	Collation finger set, format A	1 set	1 set	Wear
CP2-2010	Photo sensor, diffuse, M18	9	2	Critical
CP2-2050	Glue nozzle, 0.4 mm	4	4	Wear
CP2-2060	Glue filter element	1	2	Wear
CP2-3001	Door interlock switch with actuator	4	1	Safety
CP2-3010	Emergency stop button, complete	3	1	Safety
CP2-3100	Solenoid valve, 5/2, 24 V DC	6	1	Critical
CP2-9001	Safety label set	1	1	Safety

> **Note** Parts marked **Safety** must be replaced only with original parts of the same specification.

10.2 Ordering Parts and Service

Order parts from Acme Packaging Machinery Customer Service (fictional): parts@acme-packaging.example, +1 555 0100. Always quote the machine model, serial number and part number.

11 Decommissioning and Disposal

11.1 Decommissioning

1. Run out all products and remove remaining case blanks.
2. Empty the glue tank while hot.
3. Lock out all energy sources (Section 1.10).
4. Disconnect power, air and network connections.
5. Fit the transport locks and separate the two sections.

11.2 Disposal

Dispose of the machine in line with local regulations. Main materials: steel and stainless steel (about 85 %), aluminum, plastics, electronic components (recycle as electrical waste), and lubricants and adhesive residues (hazardous waste). Remove the PLC backup battery before disposal.

12 Appendices

12.1 Technical Data

Parameter	Value
Speed	Up to 20 cases/min
Overall dimensions (L × W × H)	5,300 × 1,600 × 2,100 mm
Machine mass	2,600 kg
Electrical supply	480 V AC, 3-phase, 60 Hz, 32 A FLA, SCCR 65 kA
Control voltage	24 V DC
Compressed air	6 bar, 350 l/min
PLC	Compact safety PLC with EtherNet/IP
Operator panel	10-inch color touch screen
Noise	74 dB(A) at operator position
Ambient temperature	5–40 °C

12.2 I/O Reference

For maintenance and electrical troubleshooting. Tag names match the PLC program and electrical drawings.

Address	Signal	Device	PLC tag
I:0/0	Emergency stop circuit OK	Safety relay	ES_Chain_OK
I:0/1	Guard doors closed	Door switches D1–D4	Doors_Closed
I:0/2	Light curtain clear	LC1	LC1_Clear
I:0/3	Start button	Operator panel	PB_Start
I:0/4	Stop button	Operator panel	PB_Stop
I:0/5	Reset button	Operator panel	PB_Reset
I:0/6	Mode switch – Automatic	Key switch	SS_Auto
I:1/0	Product present at infeed	Photo sensor B10	PE_InfeedProd
I:1/1	Case blank present	Photo sensor B20	PE_BlankPres
I:1/2	Air pressure OK	Pressure switch B30	PS_AirOK
I:1/3	Glue unit ready	Glue unit	Glue_Ready
O:0/0	Infeed conveyor run	Drive U1	R02_InfeedConv

Address	Signal	Device	PLC tag
O:0/1	Vacuum generator on	Valve Y20	SOL_Vacuum
O:0/2	Flap folder extend	Valve Y40	SOL_FlapExt
O:0/3	Stack light green	H1	LT_Green
O:0/4	Stack light red	H2	LT_Red

12.3 Control Program Cross-Reference

Shows where every rung of the control program is described in this manual. Routine and tag names match the PLC program. Rung totals are checked automatically when the manual is generated.

Routine	Rungs	Described in	Function group
MainSequence	0–3	2.11 · Modes, starting and stopping	Mode selection
MainSequence	4–11	2.11 · Modes, starting and stopping	Ready, start and cycle stop
MainSequence	12–14	2.11 · Modes, starting and stopping	Homing after reset
MainSequence	15–17	2.11 · Modes, starting and stopping	Stack light and counters
Infeed	0–5	2.11 · Infeed conveyor	Running the infeed
Infeed	6–9	2.11 · Infeed conveyor	Counting products
Infeed	10–13	2.11 · Infeed conveyor	Backup protection
PickPlace	0–7	2.11 · Collation and pick-and-place	Building the pattern
PickPlace	8–21	2.11 · Collation and pick-and-place	Pick, travel and place
PickPlace	22–24	2.11 · Collation and pick-and-place	Grip-loss recovery
PickPlace	25–27	2.11 · Collation and pick-and-place	Brake and following-error monitoring
CaseErector	0–8	2.11 · Case erector	Picking a blank
CaseErector	9–17	2.11 · Case erector	Forming and bottom-flap folding
CaseErector	18–21	2.11 · Case erector	Magazine and light curtain
GlueFlap	0–3	2.11 · Glue and flap closing	Glue readiness
GlueFlap	4–15	2.11 · Glue and flap closing	Glue pattern timing
GlueFlap	16–19	2.11 · Glue and flap closing	Compression and release
Discharge	0–9	2.11 · Case discharge	Discharge conveyor control
Alarms	0–12	2.11 · Alarms and diagnostics	Alarm detection and reset
Alarms	13–14	—	Housekeeping: Alarm history buffer and spare rung
Total	**127 of 127**		

12.4 Declaration of Conformity and Standards

The machine is designed to ISO 12100, ISO 13849-1 (PL d for guard interlocks and emergency stop), ISO 13850, ISO 14119, IEC 60204-1 and NFPA 79. A copy of the declaration of conformity is supplied with the machine documentation.

12.5 Warranty and Service

The machine carries a 12-month warranty from the date of commissioning, subject to the terms of sale. Warranty does not cover wear parts or damage from use outside the intended use. Service: service@acme-packaging.example, +1 555 0199 (24 hours).

12.6 Glossary

Term	Meaning
E-stop	Emergency stop
FLA	Full-load amperes
HMI	Human–machine interface (operator panel)
LOTO	Lockout/tagout
PLC	Programmable logic controller
RSC	Regular slotted case
Rung	One line of the control program; rung numbers are used only as references (Appendix 12.3)
SCCR	Short-circuit current rating
STO	Safe Torque Off: a drive safety function that removes motor torque